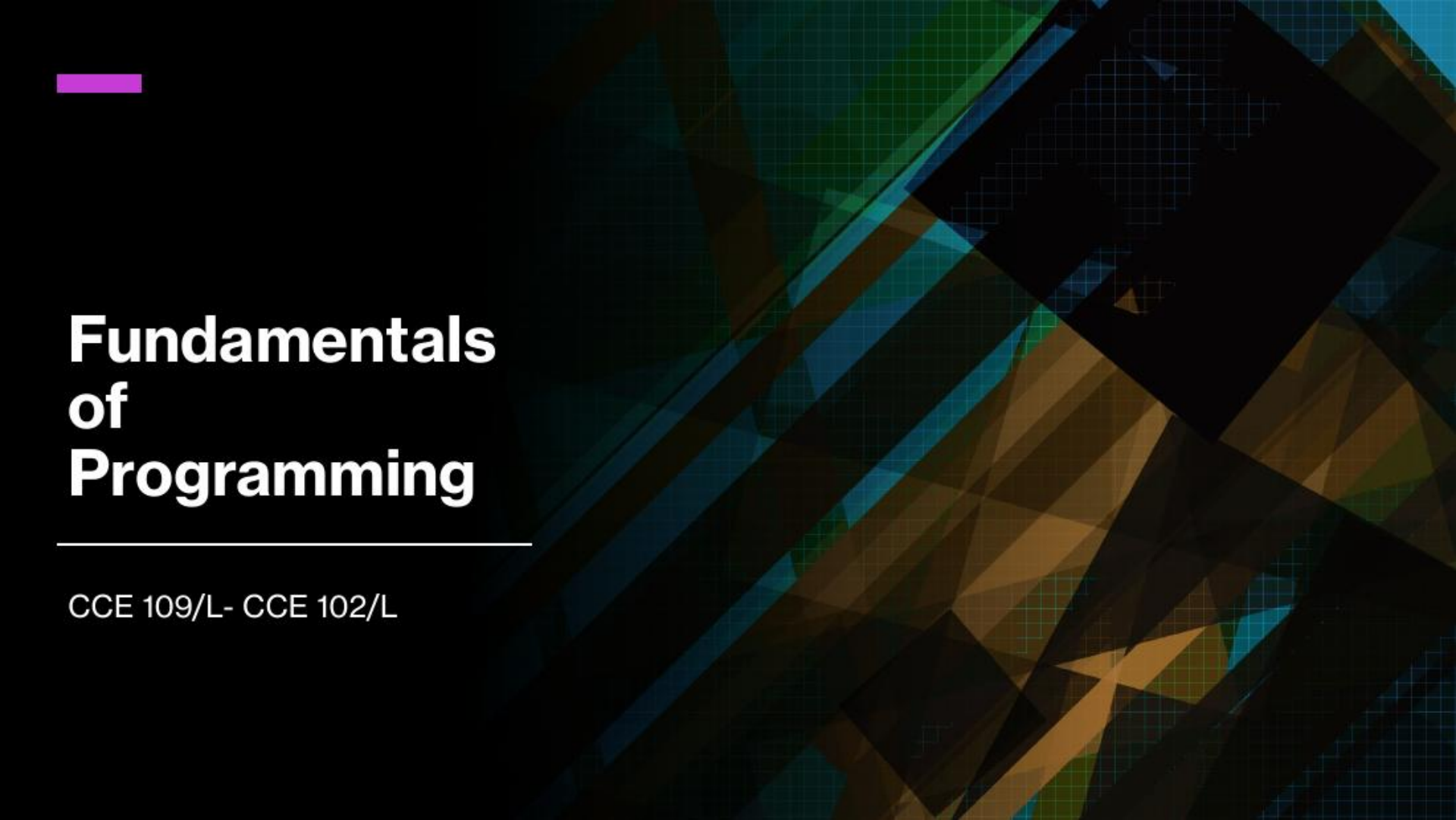




Fundamentals of Programming

CCE 109/L- CCE 102/L

What to expect from this subject:

- This subject is the foundation of programming.
- Learn the fundamentals of programming using Java as the programming language.
- Application to be used: Netbeans (for those with desktop/laptop computers)
- Any Mobile-based application where you can program using Java as programming language.



Objectives:

- Understand the basic components of a computer system: hardware and software, high-level languages and compilation, Java's place among programming languages, and the concept of an algorithm.

Computer

- A computer is an electronic device that stores and processes data. It includes both hardware and software.
- In general, hardware comprises the visible, physical elements of the computer, and software provides the invisible instructions that control the hardware and make it perform specific tasks.
- Writing instructions for computers to perform is called computer programming.

Programs

- Computer programs, known as software, are instructions to the computer, telling it what to do.
- Computers do not understand human languages, so you need to use computer languages in computer programs.
- Programming is the creation of a program that is executable by a computer and performs the required tasks.

Programming Languages

- Among the more than one hundred high-level languages, the following are well known:
 - COBOL (COmmon Business Oriented Language)
 - FORTRAN (FORmula TRANslation)
 - BASIC (Beginner's All-purpose Symbolic Instruction Code)
 - Pascal (named for Blaise Pascal)

Programming Languages

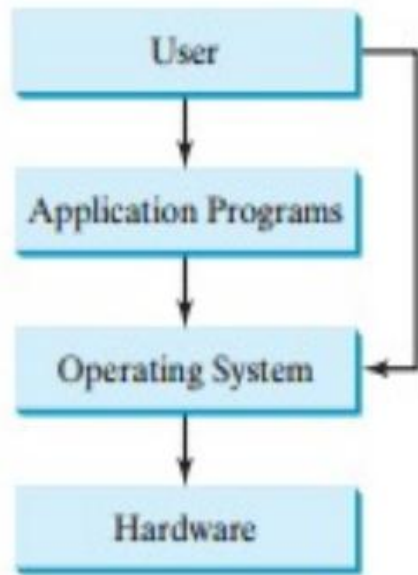
- Ada (named for Ada Lovelace)
- C (developed by the designer of B)
- Visual Basic (Basic-like visual language developed by Microsoft)
- Delphi (Pascal-like visual language developed by Borland)
- C++ (an object-oriented language, based on C)
- C# (a Java-like language developed by Microsoft)

Operating Systems

- The operating system (OS) is the most important program that runs on a computer, which manages and controls a computer's activities.
- The popular operating systems are Microsoft Windows, Mac OS, and Linux.
- Application programs, such as a Web browser or a word processor, cannot run without an operating system.
-

Operating Systems

- The major tasks of an operating system are:
 - Controlling and monitoring system activities
 - Allocating and assigning system resources
 - Scheduling operations



What is Java?

- Java is a computer programming language.
- It enables programmers to write computer instructions using English-based commands instead of having to write in numeric codes.
- It's known as a high-level language because it can be read and written easily by humans.



What is Java?

- Like English, Java has a set of rules that determine how the instructions are written.
- These rules are known as its syntax.
- Once a program has been written, the high-level instructions are translated into numeric codes that computers can understand and execute.
- Java is used to develop mobile apps, web apps, desktop apps, games and much more.

Who Created Java?

- In the early 90s, Java, which originally went by the name Oak and then Green, was created by a team led by James Gosling for Sun Microsystems, a company now owned by [Oracle](#).
- Java was originally designed for use on digital mobile devices, such as cellphones.
- However, when Java 1.0 was released to the public in 1996, its main focus had shifted to use on the internet, providing interactivity with users by giving developers a way to produce animated web pages.

Who Created Java?

- However, there have been many updates since version 1.0, like J2SE 1.3 in 2000, J2SE 5.0 in 2004, Java SE 8 in 2014, and Java SE 10 in 2018.
- Over the years, Java has evolved as a successful language for use both on and off the internet.

Why Choose Java?

- **Ease of Use:** The fundamentals of Java came from a programming language called C++.
- Although C++ is a powerful language, it is complex in its syntax and inadequate for some of Java's requirements.
- Java built on and improved the ideas of C++ to provide a programming language that was powerful and simple to use.

Why Choose Java?

- **Reliability:** Java needed to reduce the likelihood of fatal errors from programmer mistakes.
- With this in mind, object-oriented programming was introduced.
- When data and its manipulation were packaged together in one place, Java was robust.

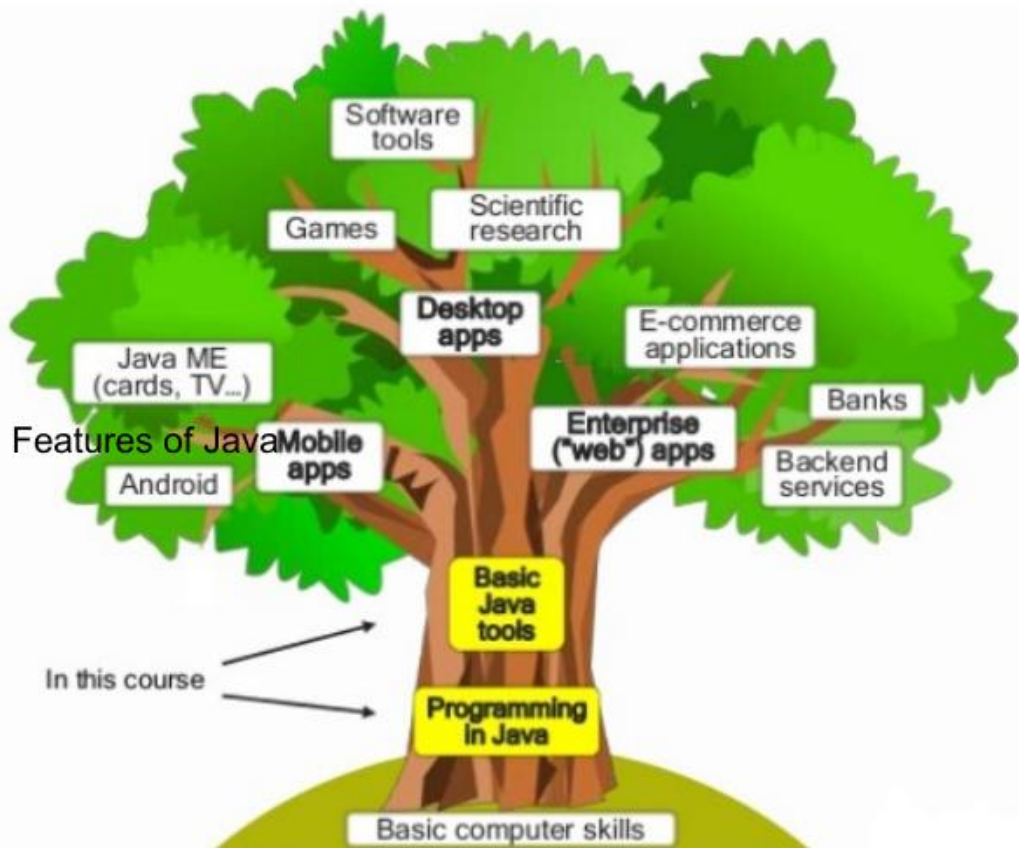
Why Choose Java?

- **Security:** Because Java was originally targeting mobile devices that would be exchanging data over networks, it was built to include a high level of security.
- Java is probably the most secure programming language to date.

Why Choose Java?

- **Platform Independence:** Programs need to work regardless of the machines they're being executed on.
- Java was written to be a portable and cross-platform language that doesn't care about the operating system, hardware, or devices that it's running on.

Java As A Tree



Features of Java



What do we need to start programming?



System Requirements:

Recommended Hardware Configurations

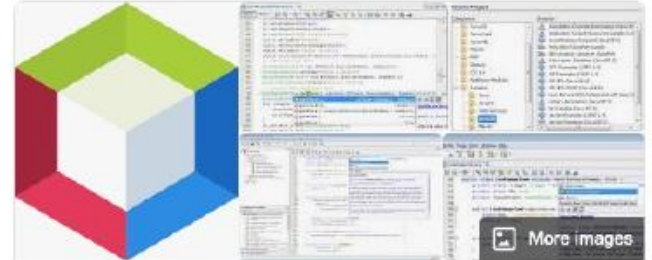
- Microsoft Windows 7 Professional/Windows 8/Windows 8.1:
Processor: Intel Core i5 or equivalent. Memory: 2 GB (32-bit), 4 GB (64-bit) ...
- Ubuntu 15.04: Processor: Intel Core i5 or equivalent. Memory: 2 GB (32-bit), 4 GB (64-bit) ...
- OS X 10.10 Intel: Processor: Dual-Core Intel. Memory: 4 GB

What do we need to start programming?

- Notepad-Text Editor
- Netbeans – this is a JAVA IDE that is open-source.
- Eclipse – also a JAVA IDE developed by the eclipse community but is already deprecated

What is Netbeans?

- NetBeans is an open-source integrated development environment (IDE) for developing with Java, PHP, C++, and other programming languages.
- NetBeans is also referred to as a platform of modular components used for developing Java desktop applications.



NetBeans

Software

NetBeans is an integrated development environment for Java. NetBeans allows applications to be developed from a set of modular software components called modules. NetBeans runs on Windows, macOS, Linux and Solaris. [Wikipedia](#)

Stable release: 12.1 (September 1, 2020; 42 days ago)

Preview release: 11.3 Beta 2 / [February 3, 2020](#); 8 months ago

Written in: [Java](#)

Operating system: [Windows](#), [macOS](#), [Linux](#), [Solaris](#); feature-limited OS independent version available

Platforms: [Java Platform](#), [Standard Edition](#), [Jakarta EE](#), [JavaFX](#)

Developed by: [Apache Software Foundation](#), [Oracle Corporation](#)

Java World Wide Web and Beyond

- Java has become enormously popular. Its rapid rise and wide acceptance can be traced to its design characteristics, particularly its promise that you can write a program once and run it anywhere.
- As stated by Sun, Java is simple, object oriented, distributed, interpreted, robust, secure, architecture neutral, portable, high performance, multithreaded, and dynamic.

Where Do I Start?

- To get started, download and install the [Java Development Kit \(JDK\)](#), and the latest [NetBeans IDE](#).
- The Java Development Kit (JDK) contains all the tools you need to compile code and run your newly written applications.
- The NetBeans IDE (integrated development environment) is an optional software utility that makes all these tools more easily accessible.

Where Do I Start?

- Download and install Netbeans.
- Download site:

<https://netbeans.apache.org/download/index.html>

How do I know if JDK is installed?

- On your command prompt window terminal type:
“java -version”

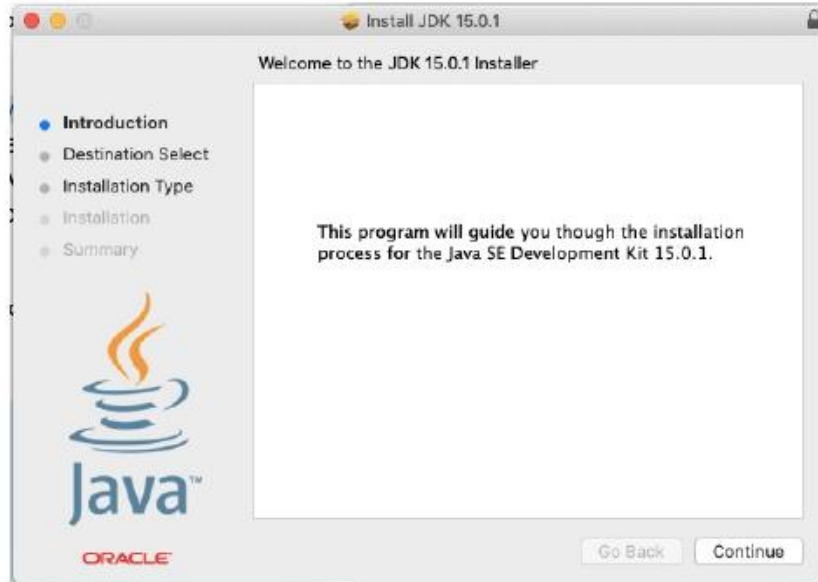
```
1 C:\>java -version
2 java version "1.6.0_18"
3 Java(TM) SE Runtime Environment (build 1.6.0_18-b07)
4 Java HotSpot(TM) Client VM (build 16.0-b13, mixed mode,
  sharing)
```

In this case, JDK **"1.6.0_18"** is installed on the computer.

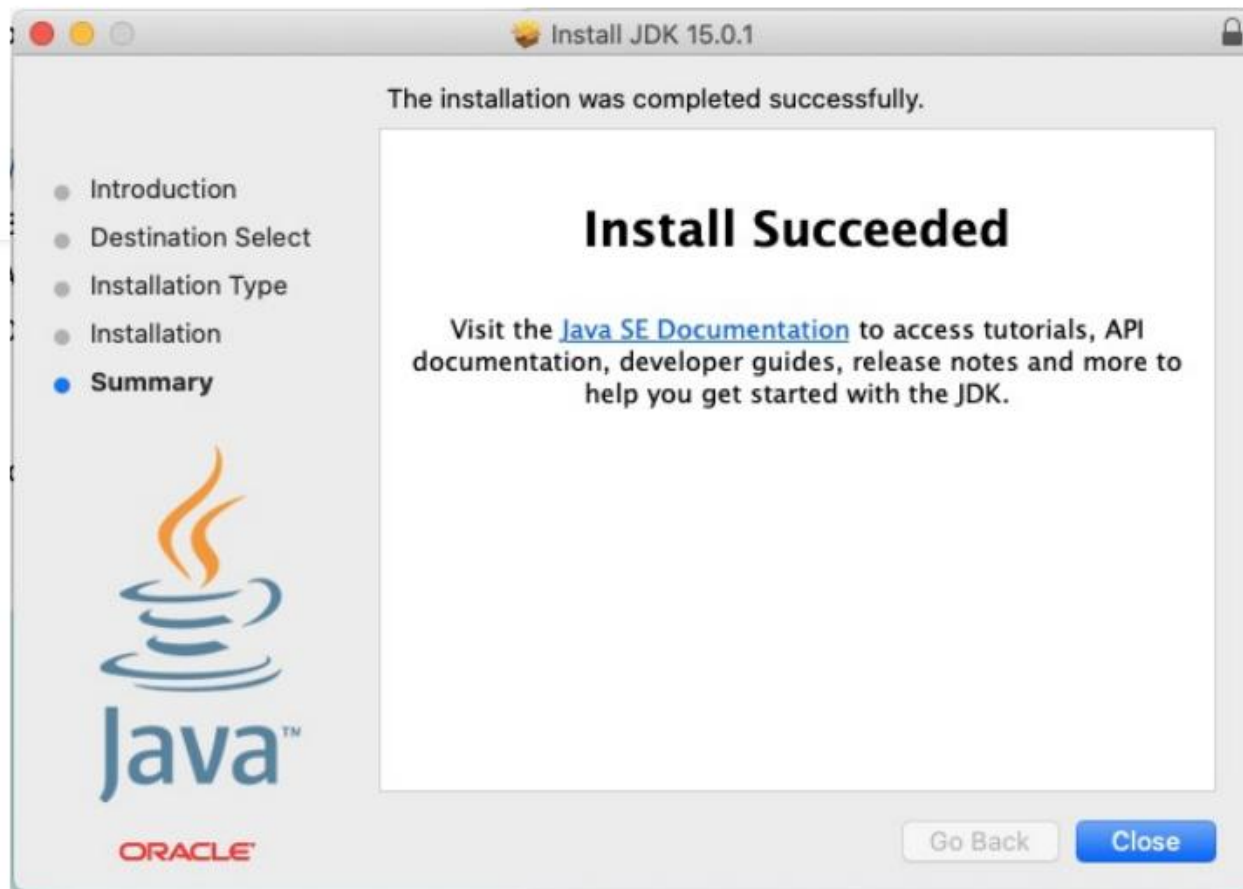
Download JDK and Netbeans

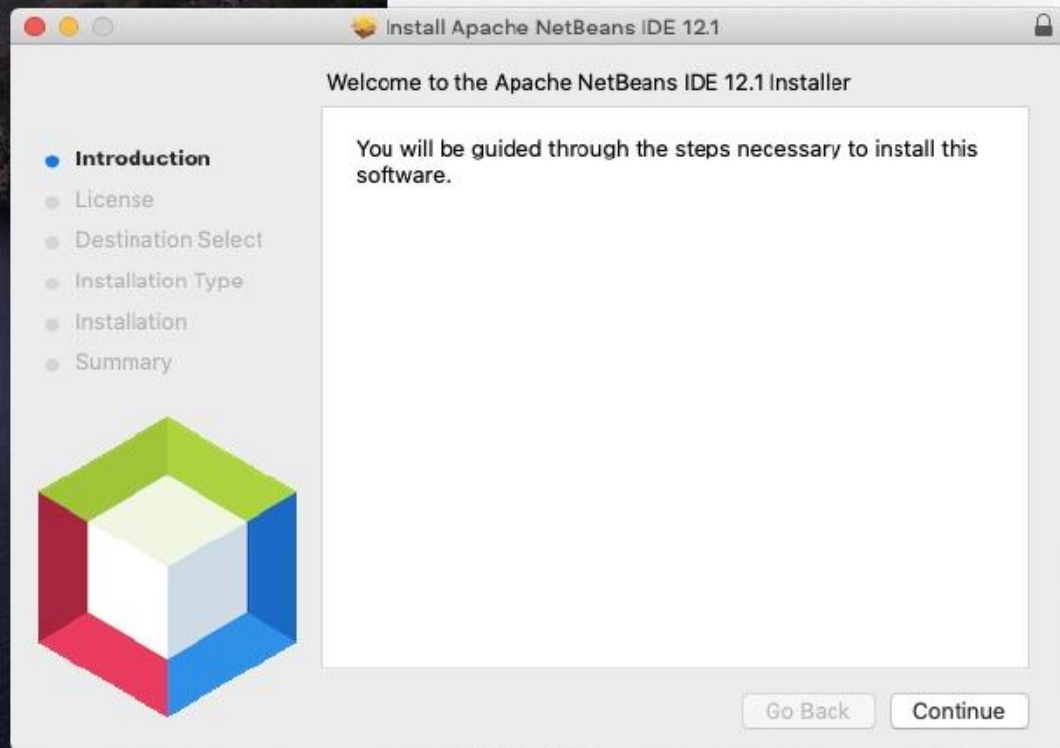
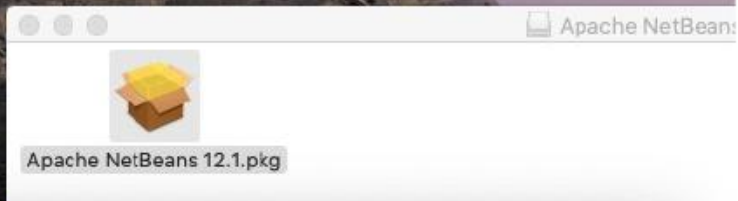
- JDK download site:
<https://www.oracle.com/java/technologies/javase-downloads.html>
- Netbeans IDE

Installing JDK



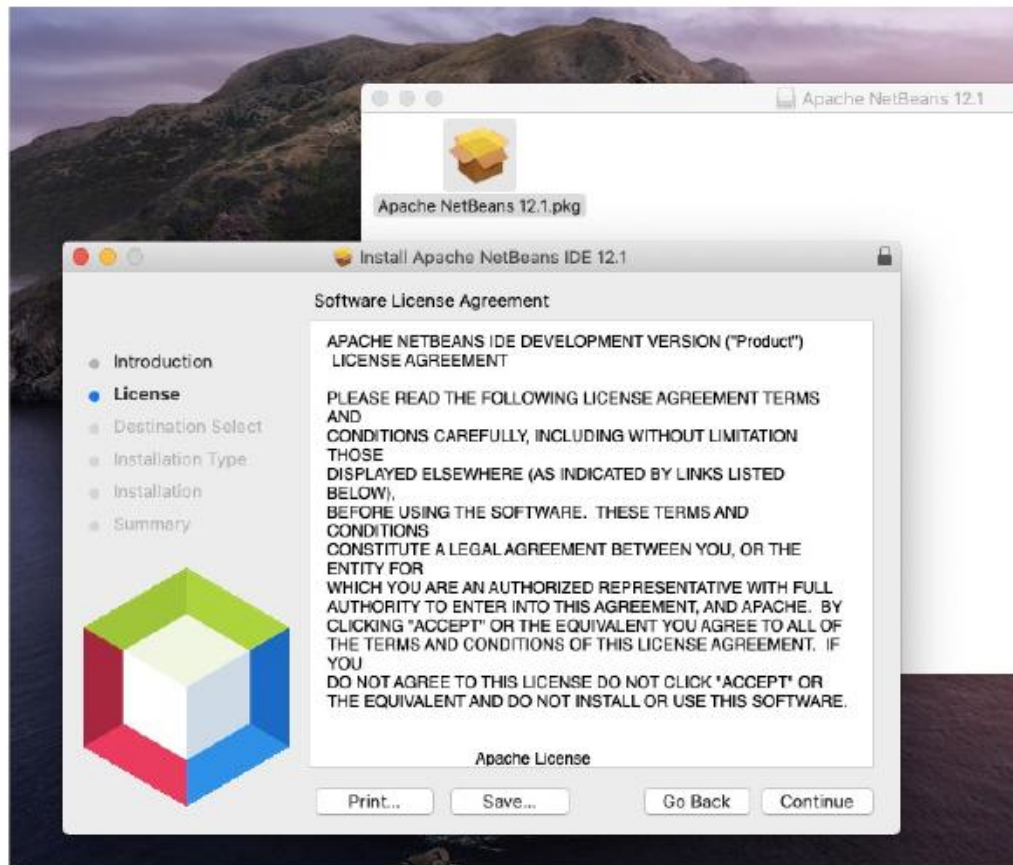


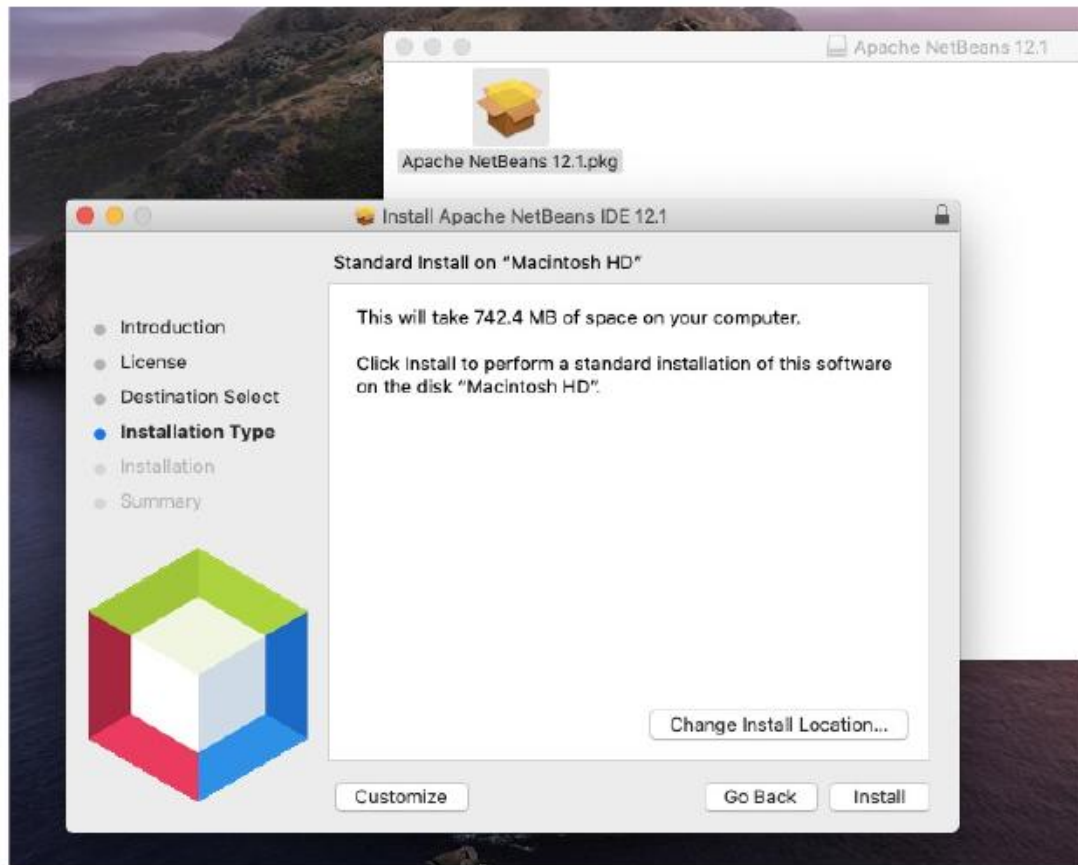




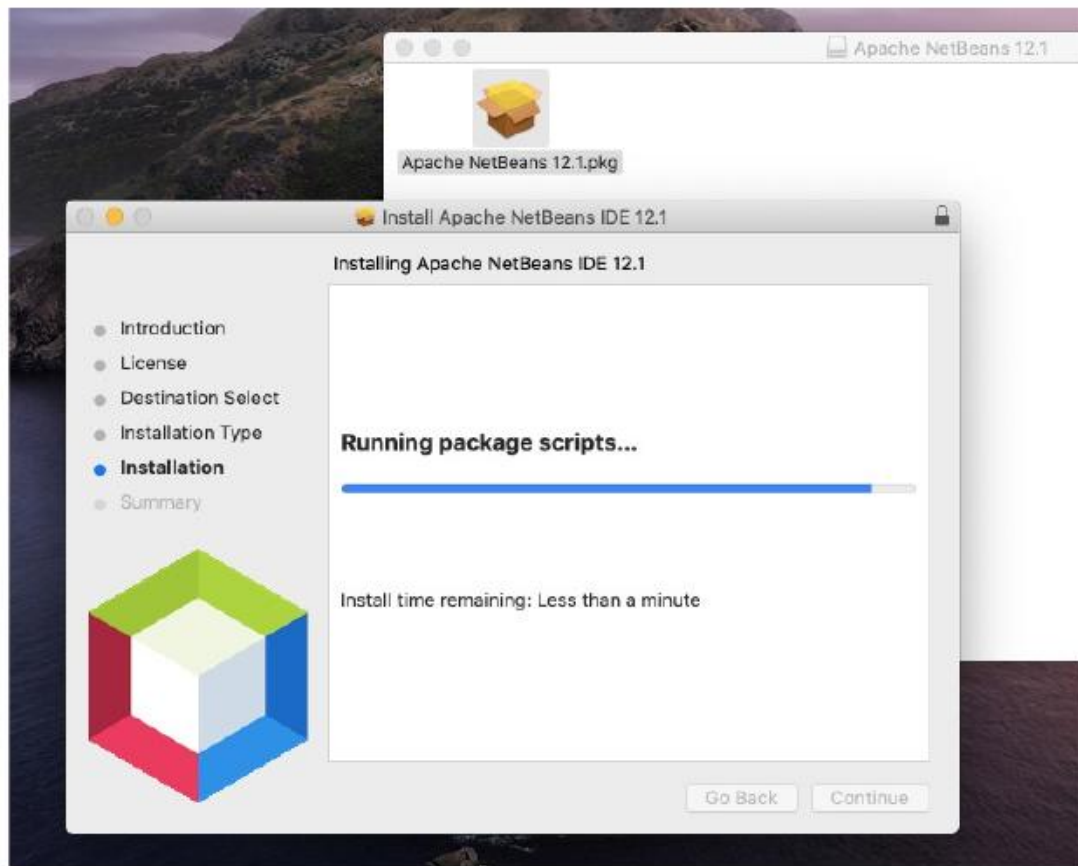
Installing Netbeans IDE

Installing Netbeans IDE

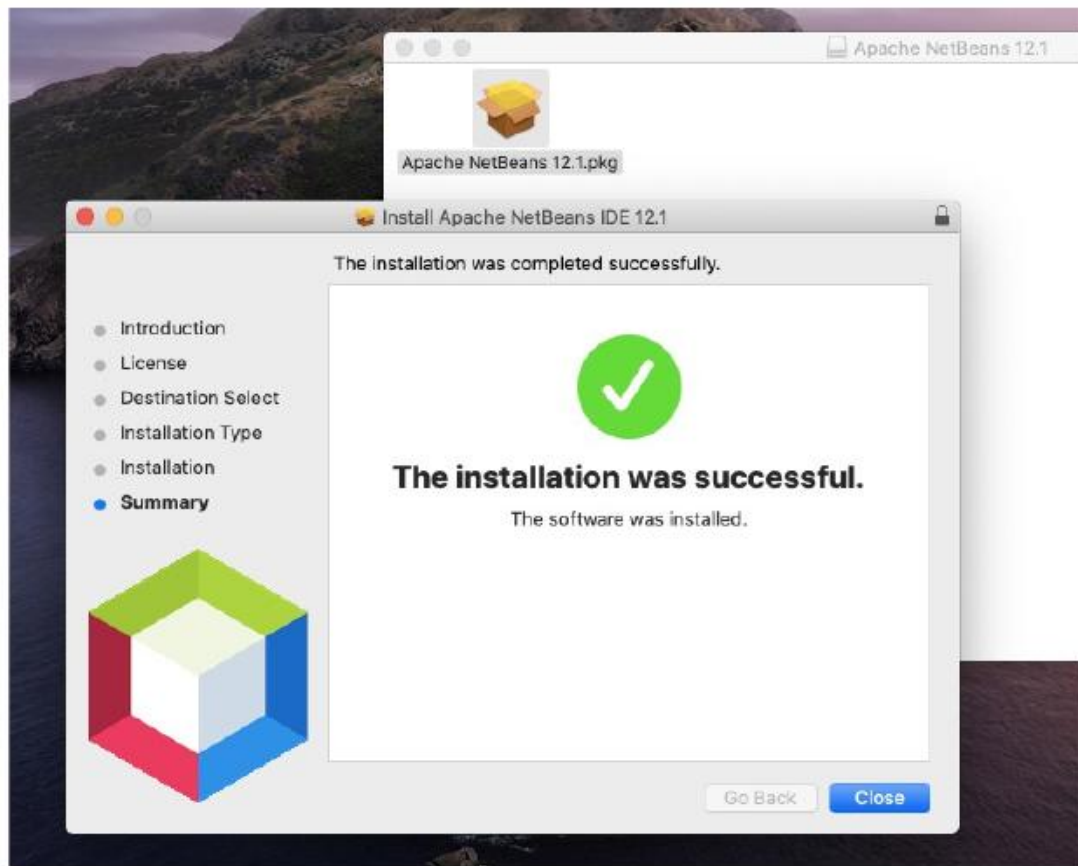




Installing Netbeans IDE



Installing Netbeans IDE



Installing Netbeans IDE

Netbeans IDE

